Blockchain without Bitcoin:
Welcome to Hyperledger Fabric

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Cloud @ Department of CSE

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Outline

- Motivation: Why Bitcoin Script Exists
- What is Bitcoin Script?
- How Script Works
- High-Level Transaction Structure
- The Stack Model (Bitcoin Script Execution)
- Standard Scripts
- p2pk: Pay-to-pubkey
- p2pkh: Pay-to-pubkey-hash
- p2sh: Pay-to-script-hash

WHY DID WE NEED THIS??

History

- The popularity of blockchain technology grew rapidly after the launch of Bitcoin, which demonstrated the potential of decentralized systems.
- As interest increased, enterprises began exploring blockchain for business applications such as supply chains, finance, and identity management.
- However, public blockchains were not suitable for enterprise needs due to SOME ISSUES LIKE..

PRIVACY
ENTERPRISES
Ye sab baatein private mein ho to behtar

SCALABILITY
ENTERPRISES
"Jor jor se bolke sabko scheme bata de"

ANONYMOUS PARTICIPANTS

ENTERPRISES

History

- However, public blockchains were not suitable for enterprise needs due to SOME ISSUES LIKE..
  - Lack of privacy – transaction data is visible to everyone.
  - Uncontrolled participation – anyone can join the network anonymously.
  - Slow transaction speed due to consensus mechanisms like Proof of Work.
  - High computational cost and energy consumption.
  - Regulatory and compliance issues for businesses handling sensitive data.

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History

- Businesses require blockchain platforms that provide:
  - Permissioned access (only authorized participants can join).
  - Identity verification for all network members.
  - High performance and faster transactions.
  - Confidential and secure transactions between organizations.
- In 2015, the Linux Foundation launched the Hyperledger project to develop enterprise-grade blockchain technologies.
- Major companies such as IBM, Intel, and SAP collaborated to build open-source enterprise blockchain frameworks and tools.

What is Hyperledger?

HYPERLEDGER

- Hyperledger is an open-source blockchain initiative designed to advance cross-industry blockchain technologies.
- It is hosted by the Linux Foundation, which supports many major open-source projects.

Hyperledger FABRIC
Hyperledger BESU
Hyperledger CALIPER
Hyperledger ANONCREDS
Hyperledger BEVEL
Hyperledger ARIES
Hyperledger SOLANG
Hyperledger INDY
Hyperledger IROHA
Hyperledger WEB3J
Hyperledger CACTI
Hyperledger FIREFLY
Hyperledger CELLO

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What is Hyperledger?

- Hyperledger focuses on enterprise-grade blockchain solutions rather than public cryptocurrencies.
- It provides frameworks, tools, and libraries for building blockchain-based business applications.
- These platforms support permissioned blockchains, where participants are known and trusted organizations.

Introduction to Hyperledger Fabric

HYPERLEDGER FABRIC

- Hyperledger Fabric is one of the most widely used frameworks in the Hyperledger project.
- It is specifically designed for enterprise blockchain applications, enabling businesses to build secure and scalable blockchain networks.
- Hyperledger Fabric supports permissioned networks, where participants are known and verified organizations rather than anonymous users.
- It provides modular architecture, allowing organizations to customize components such as consensus mechanisms and membership services.
- This makes Hyperledger Fabric suitable for industries like finance, supply chain, healthcare, and logistics.

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Key Features

- **Permissioned Network:**

  In Hyperledger Fabric, all participants are known and authenticated, ensuring secure and controlled access to the network.

- **Modular Architecture:**

  Fabric uses a flexible and modular design, allowing organizations to customize components such as membership services, consensus mechanisms, and databases.

Key Features

- **Private Communication Channels**

  Fabric supports private channels, enabling a subset of network members to conduct confidential transactions without exposing data to the entire network.

- **Smart Contracts (Chaincode)**

  Business logic in Fabric is implemented using chaincode, which functions similarly to smart contracts and automates transactions and rules.

- **Pluggable Consensus Mechanisms**

  Fabric allows different consensus algorithms to be plugged in, helping organizations choose mechanisms that provide better performance, scalability, and reliability.

Architecture Overview

The architecture of Hyperledger Fabric consists of several components that work together to process and record transactions in a secure and distributed manner.

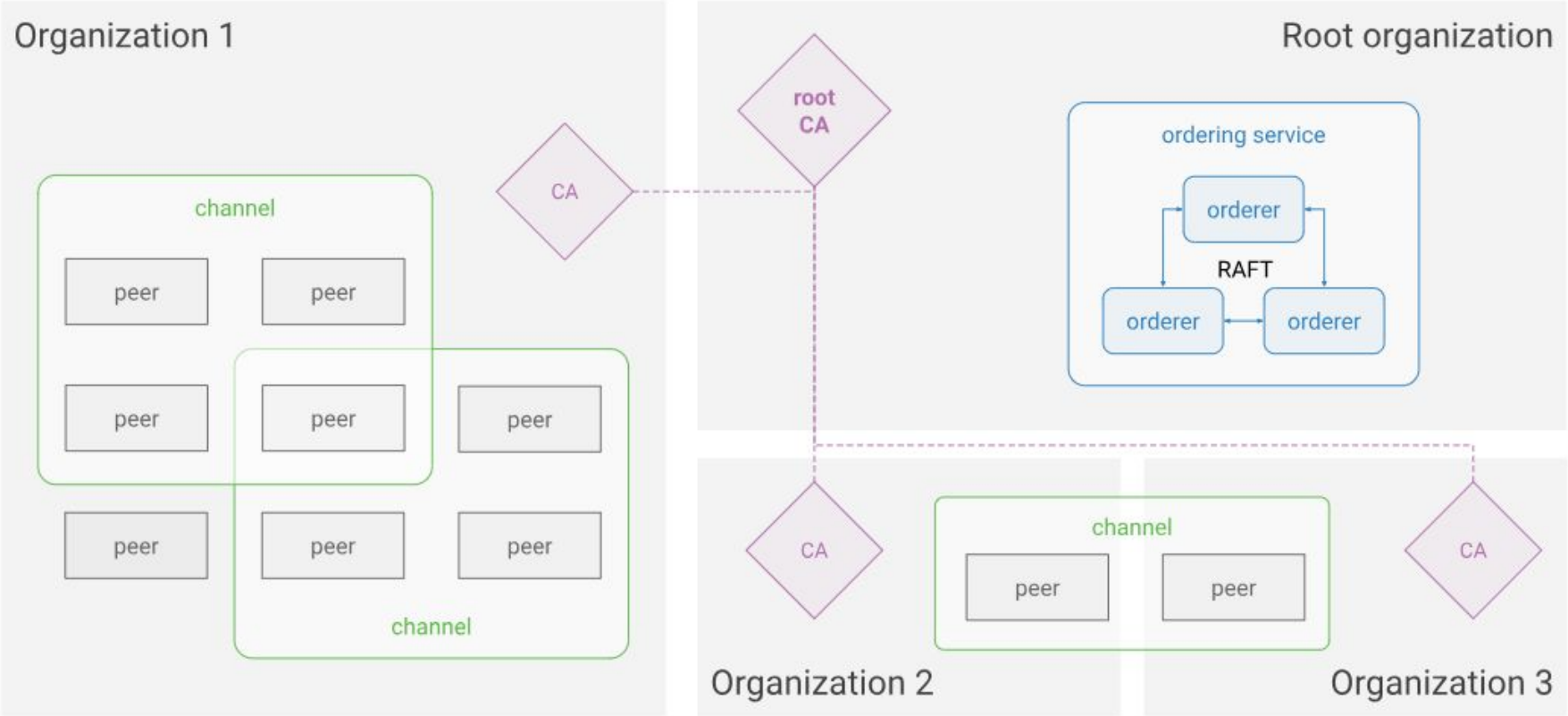
- **Client**

  The client application initiates transaction requests and communicates with the blockchain network.

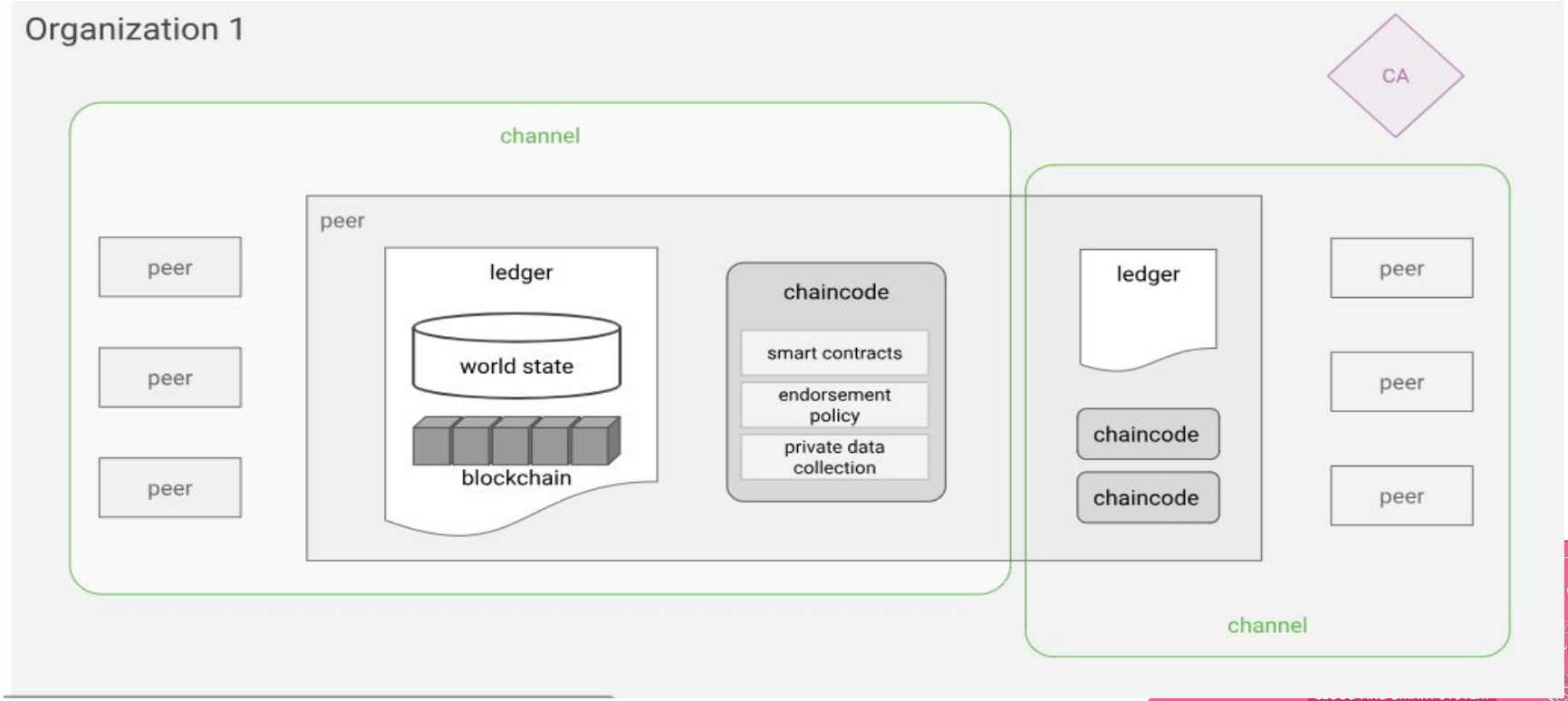
- **Peer Nodes**

  Peers are responsible for maintaining the ledger and executing chaincode (smart contracts). They also validate and endorse transactions.

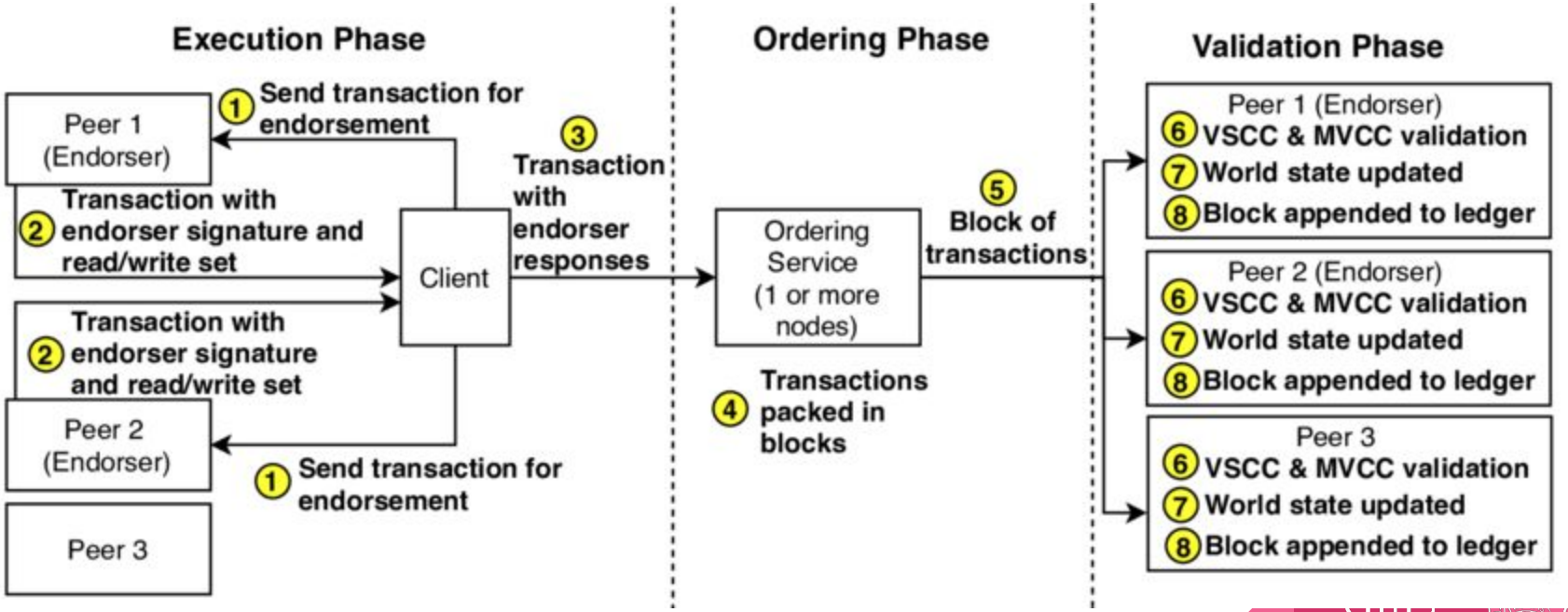
Architecture Overview(Network Topology)



Architecture Overview(Inside a Peer)



Architecture Overview



Architecture Overview

- **Ordering Service**

  The ordering service collects transactions, orders them into blocks, and distributes the blocks to peer nodes to maintain consistency across the network.

- **Ledger**

  The ledger stores the complete history of all transactions and the current state of the blockchain data.

- **Chaincode**

  Chaincode represents the business logic or smart contracts that define the rules for transactions within the network.

Core Components of Fabric

The network in Hyperledger Fabric is structured using several key components that enable organizations to securely collaborate on a shared blockchain system.

- **Organizations**

  Organizations represent different participants or companies in the blockchain network. Each organization manages its own peers and users.

- **Peers**

  Peer nodes maintain the ledger and execute chaincode (smart contracts). They also validate and endorse transactions before they are added to the blockchain.

Core Components of Fabric

- **Orderers (Ordering Service)**

  Orderer nodes are responsible for collecting transactions, ordering them into blocks, and distributing these blocks to peers to maintain a consistent blockchain across the network.

- **Channels**

  Channels allow a subset of network members to create a private blockchain where only authorized participants can view and process specific transactions.

- **Certificate Authority (CA)**

  The Hyperledger Fabric CA issues digital certificates and manages identities for users and components in the network, enabling secure authentication and permissioned access.

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Core Components of Fabric

World state

- This is a key value store that contains data created by transactions from the blockchain. Unlike the blockchain, the world state is mutable. It represents the latest values for all keys updated in the transactions.
- Even if a smart contract "reads" and "saves" the world state data, during the smart contract execution, there is no actual change of the world state. A smart contract always has access to the world state from the previous block and the result of execution contains a read/write set, i.e. a list of keys read and a list of keys and values to be updated.

Transaction Flow

Hyperledger Fabric uses a unique **Execute → Order → Validate** transaction model, which improves scalability and performance compared to platforms like Ethereum.

1. **Execute (Endorsement Phase)**: The client sends a transaction proposal to endorsing peer nodes. These peers execute the chaincode to simulate the transaction and generate a response without updating the ledger.
2. **Order (Ordering Phase)**: The endorsed transactions are sent to the ordering service, which orders transactions and groups them into blocks.
3. **Validate (Validation Phase)**: Peer nodes validate the transactions by checking endorsement policies and ensuring no conflicts occurred. Valid transactions are then committed to the ledger.

Identity and Security

Hyperledger Fabric provides strong identity and security mechanisms, ensuring that only authorized participants can access and interact with the network.

- **Membership Service Provider (MSP):** The MSP defines the rules for identity validation in the network. It verifies the identities of participants and determines whether they are allowed to join and perform transactions.
- **Certificate Authority (CA):** The Hyperledger Fabric CA is responsible for issuing and managing digital certificates for users, peers, and organizations in the network.
- **Identity Management:** Each participant in the network has a unique digital identity based on cryptographic certificates, enabling secure authentication and accountability.

Smart Contract-Chaincode

In Hyperledger Fabric, smart contracts are implemented as Chaincode, which encapsulates the business logic of transactions.

- **Business Logic**

  Chaincode defines the rules and procedures for how assets are created, transferred, or updated within the network, ensuring consistent and automated execution.

- **Supported Programming Languages**

  Developers can write chaincode using Go, Java, or Node.js, allowing flexibility to use familiar languages for enterprise applications.

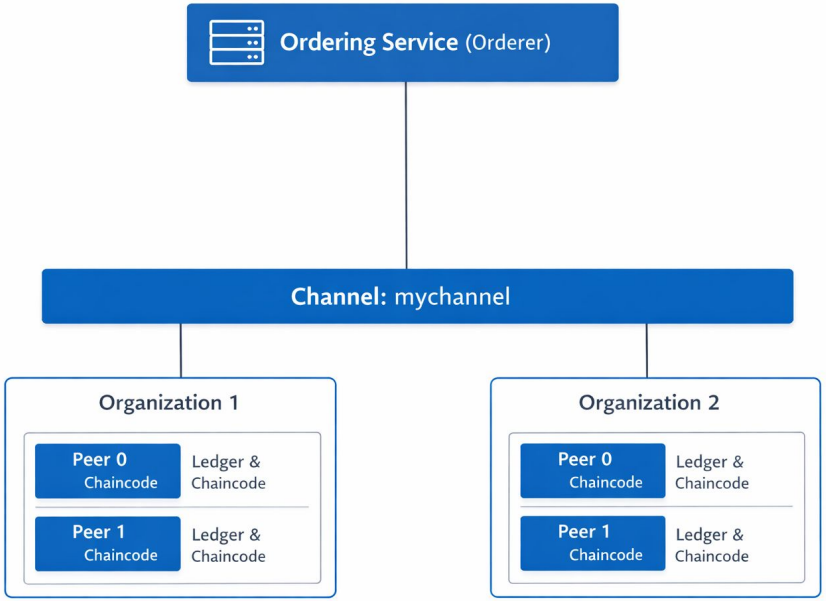
Smart Contract-Chaincode

- **Interaction with Ledger**

  Chaincode interacts directly with the ledger, enabling it to read the current state, update records, and log transaction history securely.

By using chaincode, organizations can automate processes, enforce rules, and ensure that all transactions follow predefined conditions, while maintaining the security and integrity of the blockchain network.

Hyperledger Fabric Network



Hyperledger Fabric Network

A basic Fabric network consists of multiple organizations, each managing its own peers. This structure illustrates how Fabric enables multiple organizations to participate in a permissioned network while keeping control and privacy within each entity.

- **Organization 1 → Peer Nodes**

  Maintains a set of peer nodes that store the ledger and execute chaincode.
- **Organization 2 → Peer Nodes**

  Another organization with its own peers, collaborating on the network while maintaining separate control.

Hyperledger Fabric Network

- **Ordering Service**

  A centralized service that orders transactions into blocks and ensures consistency across all peers.

- **Channel Connecting Them**

  A channel provides a private communication layer, allowing the organizations to share ledger updates and transaction data securely.

Demonstration

https://hyperledger-fabric.readthedocs.io/en/release-1.4/build_network.html

Applications

- **Supply Chain Management:** Enables traceability, transparency, and secure tracking of goods across multiple organizations.
- **Banking and Finance**: Used for secure transactions, payment settlements, and trade finance, reducing intermediaries and improving efficiency.
- **Healthcare**: Facilitates secure sharing of medical records while maintaining patient privacy and compliance with regulations.
- **Digital Identity**: Provides verifiable, tamper-proof digital identities for individuals and organizations.
- **IoT Networks:** Helps manage secure device interactions and data exchange in Internet of Things ecosystems.

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Advantages

- **High Scalability**: Fabric's Execute → Order → Validate model and modular design allow networks to handle large volumes of transactions efficiently.
- **Privacy and Confidentiality**: Supports permissioned networks and private channels, ensuring sensitive data is only shared with authorized participants.
- **Modular Design**: Components like consensus, membership services, and ledger storage are pluggable and customizable, allowing flexibility for different enterprise needs.
- **Enterprise-Ready Infrastructure**: Built for robust, secure, and compliant business applications, Fabric integrates easily with existing enterprise systems and supports multi-organization collaboration.

“Bhaisahab, aap convince ho gaye ya main aur bolun?"

Thank You

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https://souravkaddya.in/

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